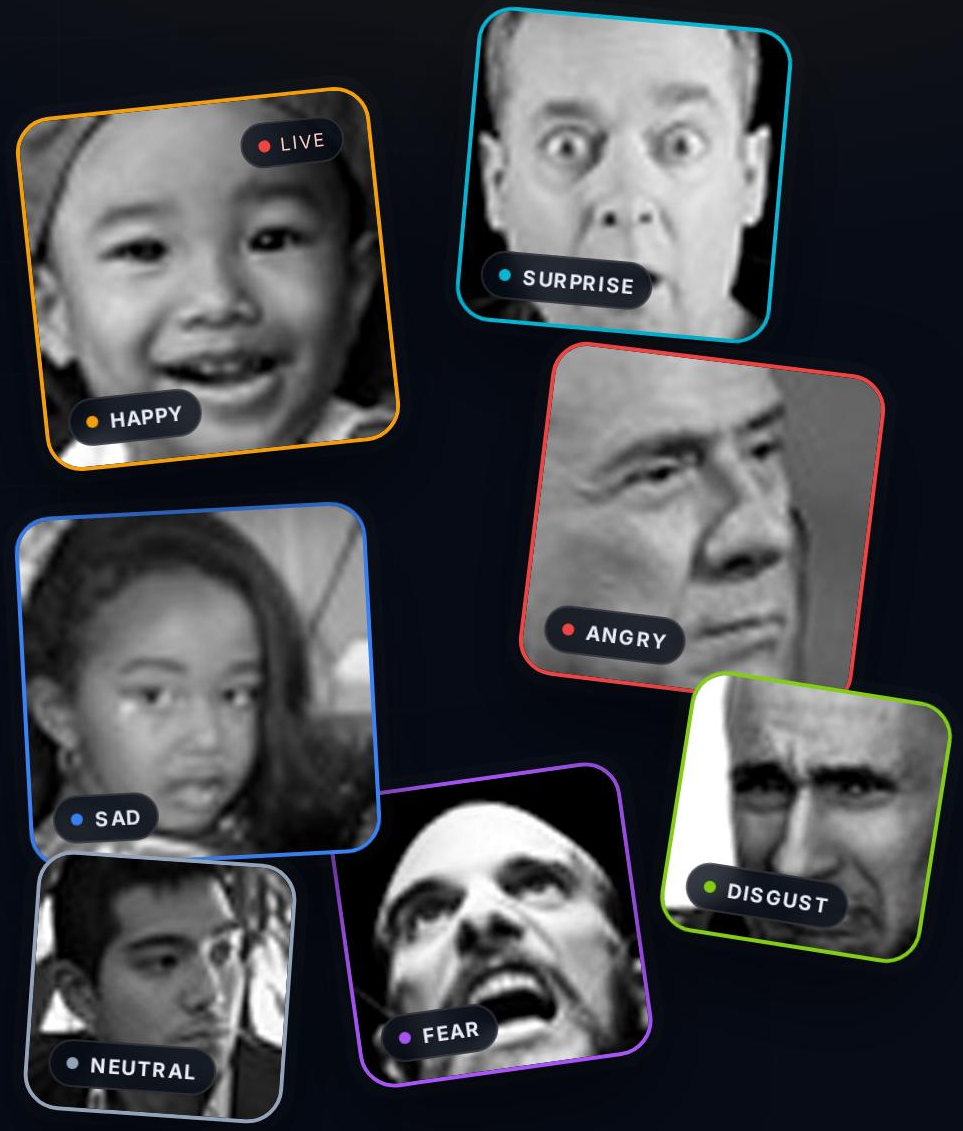
DEEP LEARNING · COMPUTER VISION

Reading faces in real time.

A compact CNN that turns a webcam frame into one of seven expressions — trained from scratch on FER2013 and explained with Grad-CAM.

35,887 faces | **7** expressions | **1.77M** parameters | **CPU** real-time



THE PROBLEM

One glance, seven labels.

Faces are one of the densest social signals we produce. A camera and a small CNN can classify an expression in milliseconds — with real limits.

Human–computer interaction

Interfaces that notice frustration, confusion or delight without waiting for a click.

Accessibility & health

Screening aids, affect-aware tutoring and communication tools for people who cannot speak.

Media & retail

Audience reactions, content testing and responsive displays — if consent and bias are handled.

One caveat up front: a facial expression is not a direct readout of an inner feeling. We will come back to this.

THE DATA

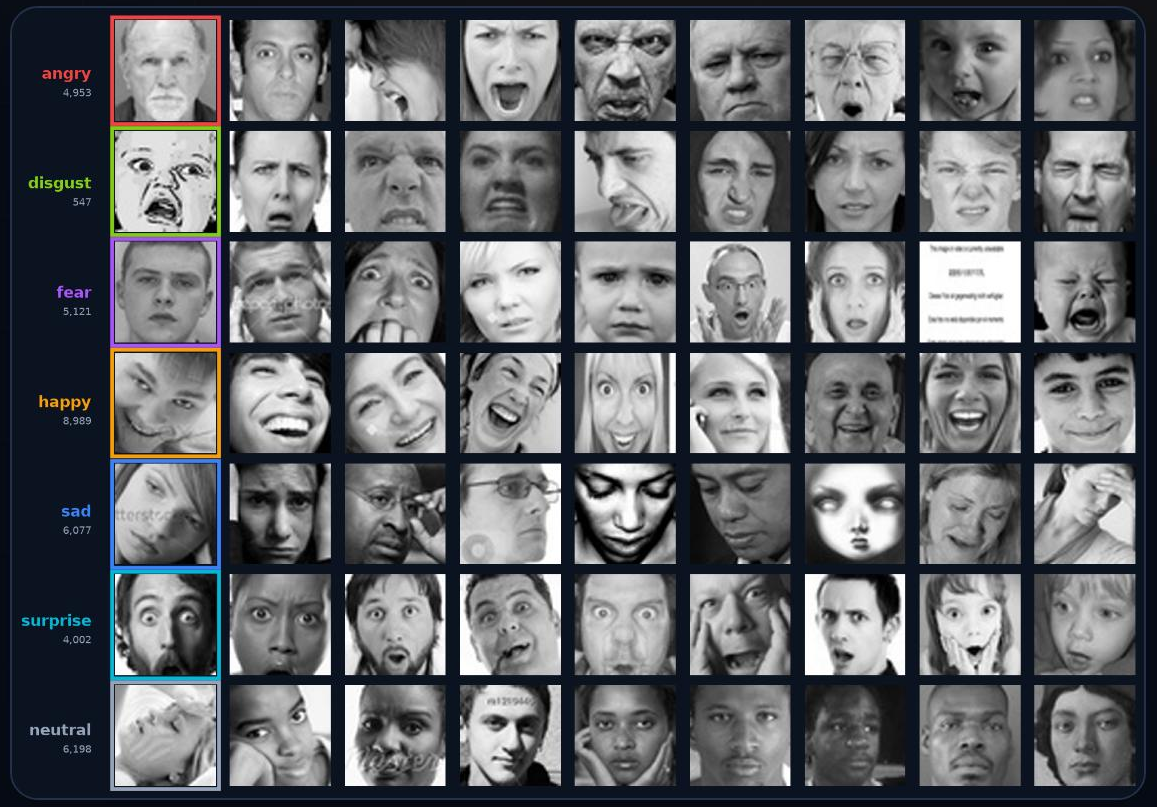
Tiny faces, messy labels.

FER2013 collects 35,887 grayscale web images, cropped to 48×48 and labelled with one of seven expressions. It is small, noisy, and wonderfully hard.

35,887 IMAGES	**48×48** GRAYSCALE
28,709 TRAIN	**7,178** PUBLIC TEST

- disgust is **16×** rarer than happy

10% of train is held out for validation, stratified by class.



WHY IT IS HARD

The model sees 1,024 pixels. The label sees a whole person.



- **Ambiguity is built in**

  “Sad” and “neutral” share a mouth. “Fear” and “surprise” share wide eyes. Human annotators disagree, and so does the model.

- **Low resolution, no context**

  48×48 grayscale removes colour, pose, body language and the situation. The model must infer everything from local geometry.

- **Labels are noisy and dated**

  Web-scraped images from 2013, labelled by crowdworkers. Cultural display rules are not represented equally.

> “Disgust is rare, visually close to anger, and often mislabelled by humans too.”

547

THE SYSTEM

From webcam frame to expression in under a blink.

Classical face detection finds the crop. Everything after that is one small neural network and a smoothing filter.

Frame

Webcam, video file or a folder of images at 640–960 px.

›

Detect

Haar cascade returns the largest frontal face box.

›

Crop & normalise

Pad 10%, convert to grayscale and resize to 48×48.

›

CNN

Seven 3×3 convolutions, batch norm, dropout, global classifier.

›

Softmax

Seven probabilities; top-1 drives the label and confidence.

›

Smooth

Average the last seven frames so labels do not flicker.

End-to-end: detect, classify and draw in real time on a laptop CPU.

THE MODEL

A small VGG-style CNN, built for 48-pixel faces.

Four convolutional stages, one dense head. No pretraining, no landmarks, no external data.

48×48

Grayscale face
normalised to [−1, 1]

→

24×24

2× conv 3×3
BN + ReLU
max-pool 2×2
dropout 0.25

→

12×12

2× conv 3×3
BN + ReLU
max-pool 2×2
dropout 0.25

→

6×6

2× conv 3×3
BN + ReLU
max-pool 2×2
dropout 0.30

→

3×3

1× conv 3×3
BN + ReLU
max-pool 2×2
dropout 0.30

→

7

flatten 2,304
FC 512 + BN + ReLU
dropout 0.40
FC 7

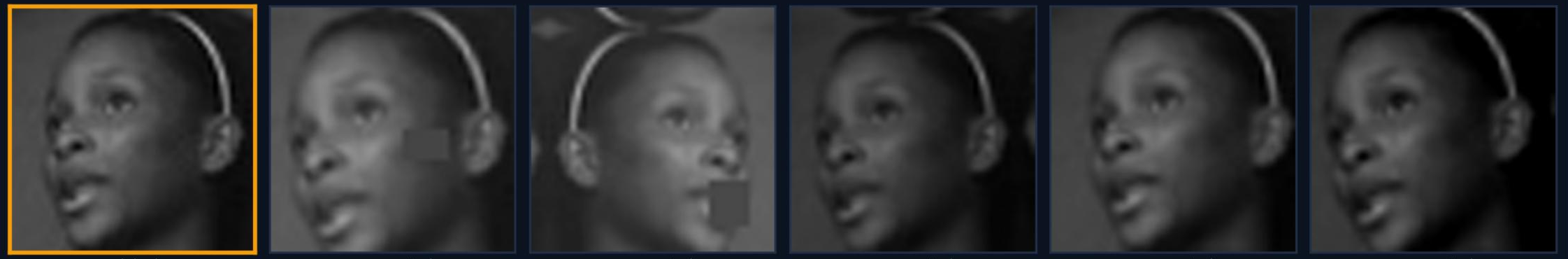
1.77M	7	6×6×256	~11.1 ms
TRAINABLE PARAMETERS	CONVOLUTION LAYERS	GRAD-CAM TARGET	SINGLE-FACE CPU INFERENCE

TRAINING

Make the most of 28,709 faces.

Random flips, rotations and erasing turn one face into many. Class weights stop the majority from winning.

One face, many plausible variations



original · augmented · augmented · augmented · augmented · augmented

flip · ±12° rotation · 0.9–1.1× scale · ±20% brightness/contrast · random erasing

OPTIMISER **AdamW · 1e-3**	SCHEDULE **cosine · 3 ep warm-up**	LOSS **CE + label smooth 0.05**	IMBALANCE **√-inverse class weights**
BATCH **128**	REGULARISATION **dropout 0.25–0.40 · wd 1e-4**	EPOCHS **36 epochs · early stop 10**	COMPUTE **57 min · CPU only**

No shortcuts: trained from random initialisation on FER2013 only — no ImageNet weights, no face landmarks.

RESULTS

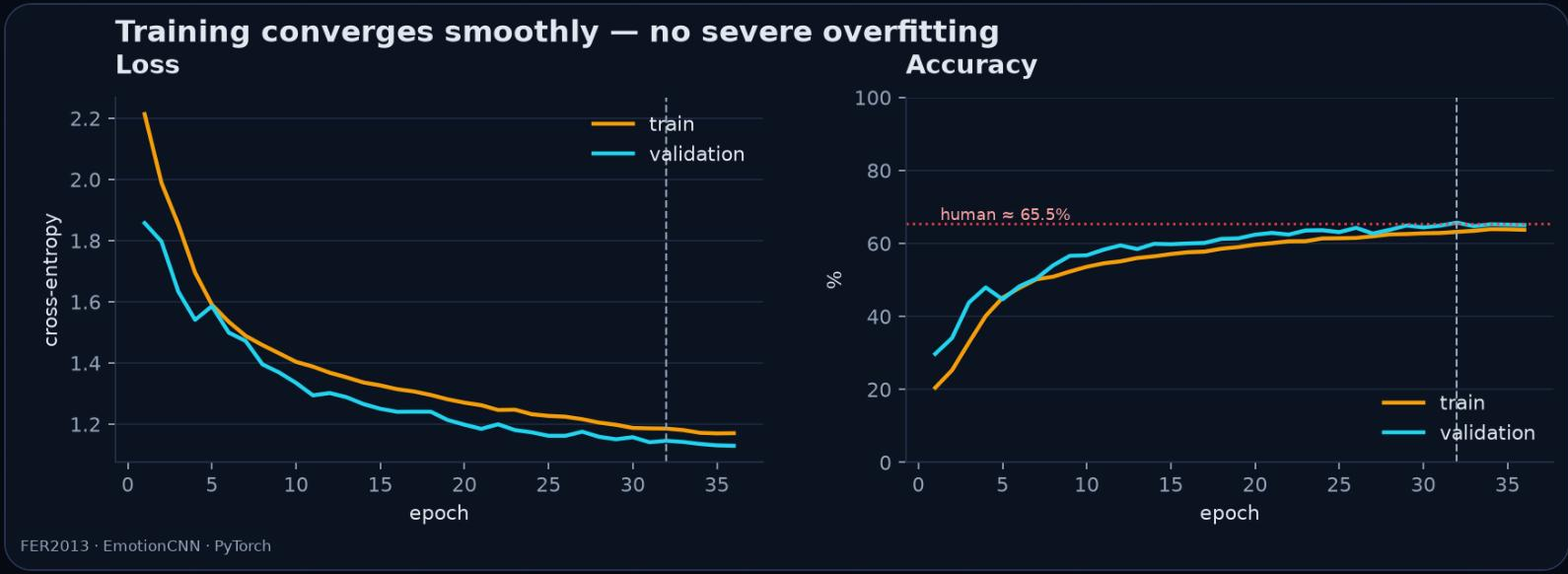
65.0% on faces it has never seen.

Held-out FER2013 test split · 7,178 images · flip TTA · -0.5 pts vs the original human benchmark.

65.0%
TEST ACCURACY
4,664 of 7,178 correct

61.5%
MACRO F1
unweighted mean over 7 classes

1.77M
PARAMETERS
11.1 ms per face · CPU



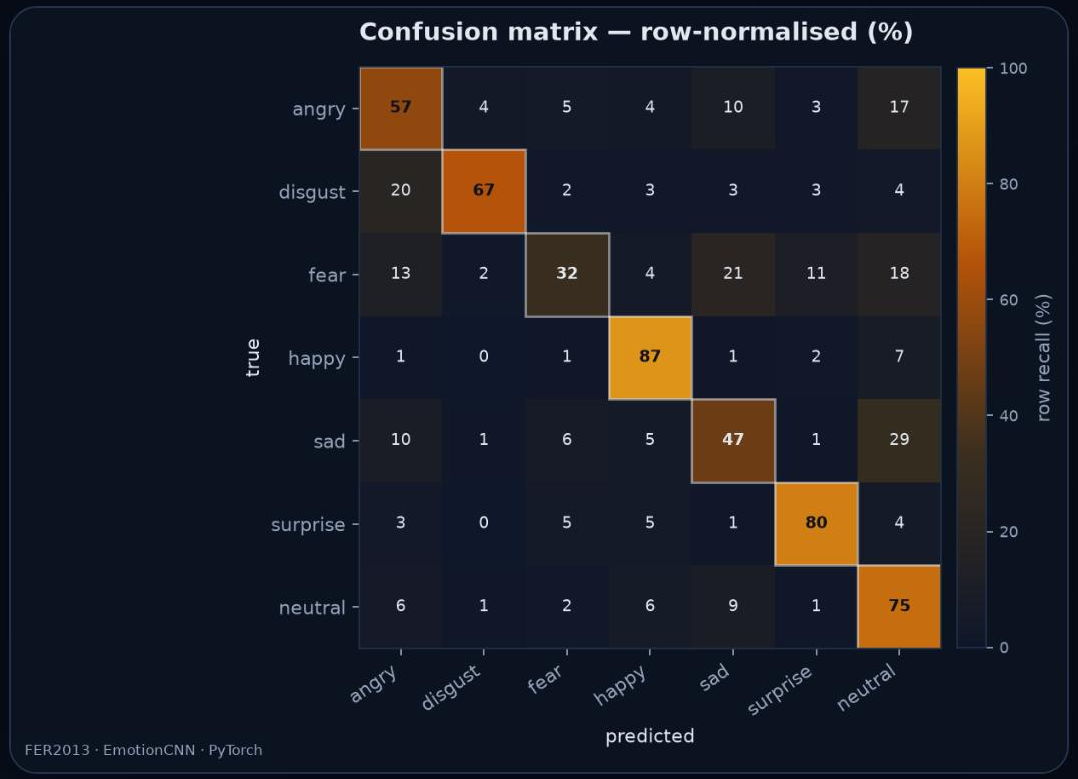


Best validation epoch: 32. Final train–validation gap: 1.3 points.

WHERE IT FAILS

The mistakes are the interesting part.

Rows are true labels. Bright diagonal is good; bright off-diagonal cells are systematic confusions.



Happy is the easy win

A wide mouth and raised cheeks are hard to confuse. Happy recall reaches 87%.

The big confusion: sad → neutral

29% of all sad faces are predicted as neutral. The mouth shape overlaps and the crops are tiny.

Disgust is starved

Only 547 training examples and 111 test examples, visually close to anger. Recall is 67%.

Takeaway: aggregate accuracy hides which faces fail. Always inspect the matrix.

PER-CLASS VIEW

Accuracy is a mean. Users experience classes.

Precision, recall and F1 for each expression on 7,178 test faces.



61.5%

MACRO F1

every class counts equally

Strong: happy · surprise

Distinct geometry and enough examples. These are the classes a demo should show first.

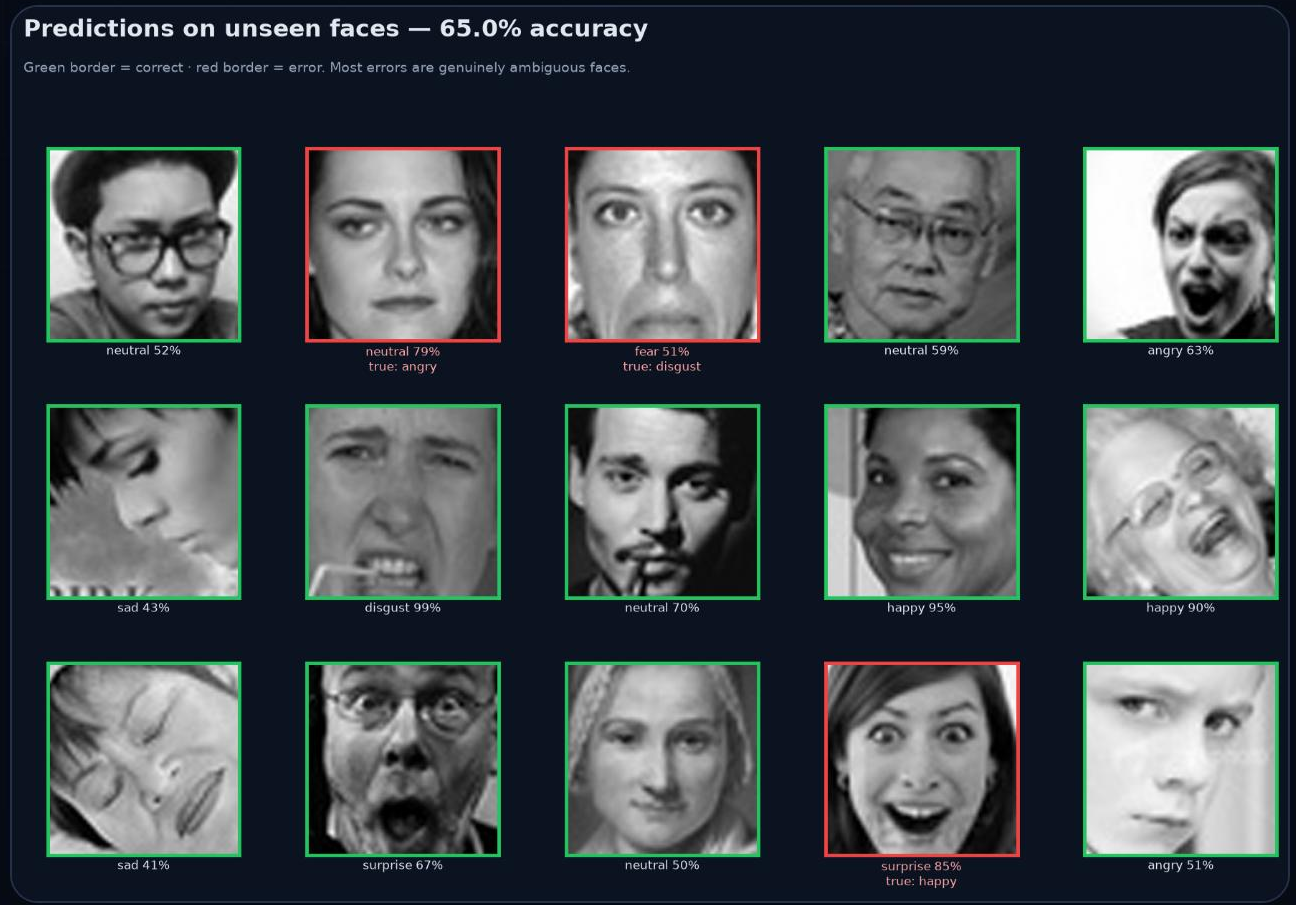
Weak: disgust · fear

Rare, subtle, and easily absorbed into anger or sadness. 3.5 points separate accuracy from macro F1.

GALLERY

Unseen faces, real mistakes.

A random sample of 15 test images. Green border = correct, red = wrong. Confidence is the model's softmax, not a guarantee.



The model handles clean, frontal faces well. When it fails, the face is often low-contrast, occluded, or genuinely between two labels.

> **Read the red boxes:** a confident wrong answer is more dangerous than a low-confidence one.

held-out only · no cherry-picking · flip TTA

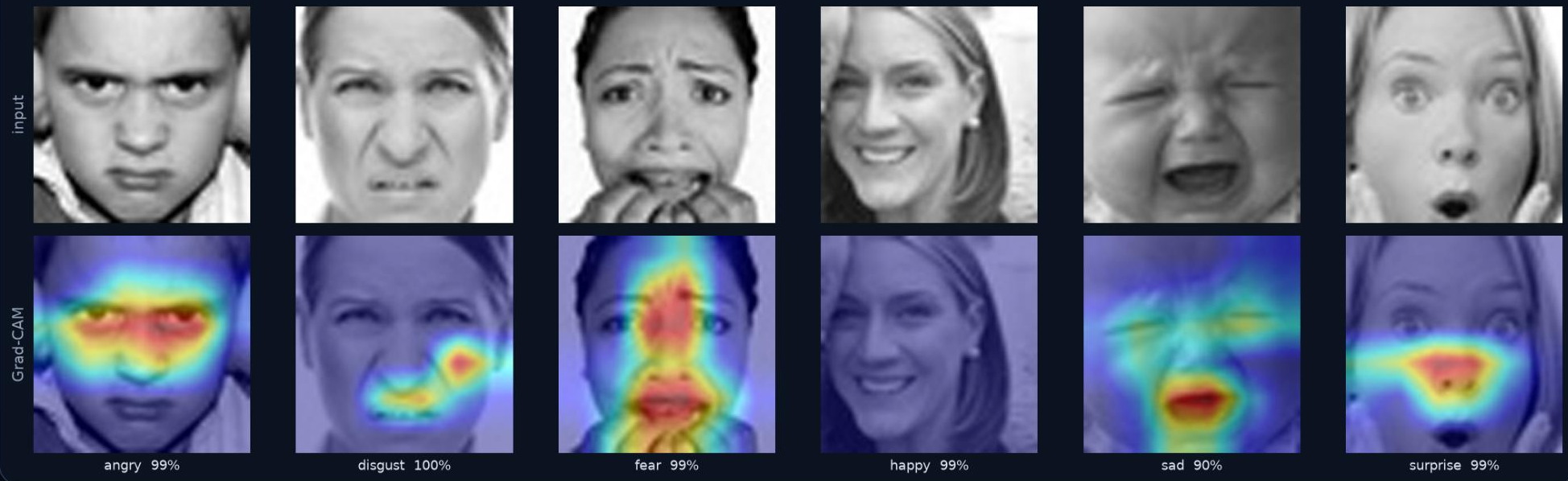
EXPLAINABILITY

It learned eyes and mouths, not identities.

Grad-CAM highlights the pixels that most increase the predicted class score.

Where the CNN looks — Grad-CAM

Warm regions drive the prediction: eyes, brows and mouth shape — not background or hair.



Focus on the face
Heat sits on brows, eyes and mouth.

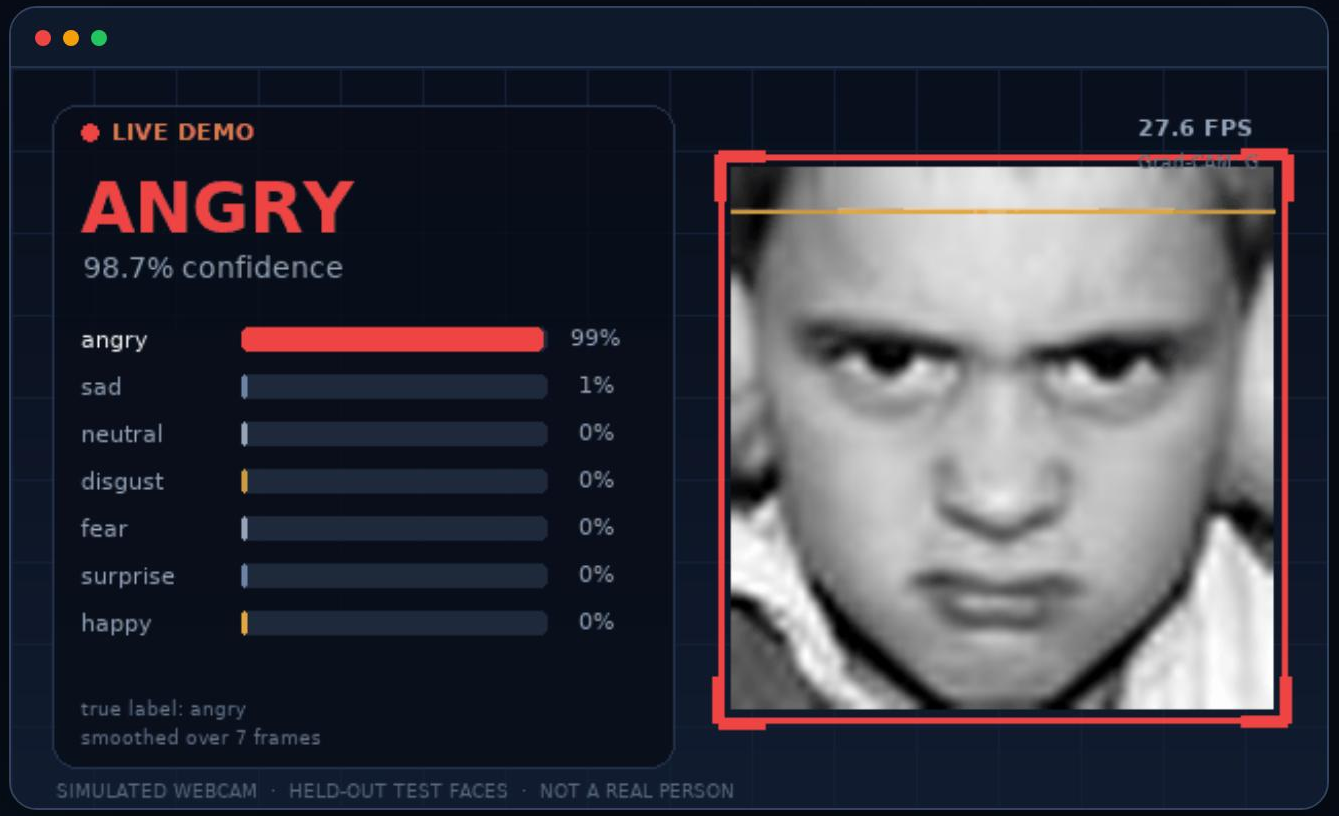
Class-specific cues
Happy lights up the mouth; fear the eyes.

Useful debugging
Border heat means a dataset artefact.

LIVE SYSTEM

Real time, one face at a time.

OpenCV handles capture and detection; PyTorch runs the CNN. The overlay shows the top-1 label, all seven probabilities and the smoothed confidence.



Simulated from held-out test faces — no real person in this recording. Run the same code on a webcam with .

Detect & crop
Haar cascade picks the largest face; 10% margin keeps the chin and brow in frame.

Classify with flip TTA
Average the face and its mirror image before softmax for a small accuracy gain.

Smooth over 7 frames
Prevents label flicker while still reacting in about a quarter of a second.

Explain on demand
Press G for Grad-CAM, B for bars, S for a screenshot, Q to quit.

CNN forward 11.1 ms per face · 29.6 FPS full loop on this CPU.

LIMITS & ETHICS

What this model must not be used for.

A 48×48 classifier is a demo, not a diagnostic. The gap between expression and feeling is where harm happens.

Lighting, pose & occlusion

Side profiles, masks, glasses and harsh shadows change the crop. Real webcams are not curated like FER2013.

Culture & context

Display rules differ across cultures, ages and contexts. A smile can signal politeness, stress or disagreement.

Dataset bias

Web images from 2013, crowd labels, no demographic balance. Performance will not transfer equally.

Privacy & consent

Emotion data is sensitive. Capture needs informed consent, retention limits and a clear opt-out.

A facial expression is not a feeling.

Use this system to explore computer vision, not to judge people. Never use it for hiring, policing, credit, or medical decisions.

Deployment rule: if a wrong prediction could change someone's access, safety or dignity, this model is not ready.

TAKEAWAYS

Small model. Real time. Honest numbers.

A compact CNN is enough

1.77M parameters, trained from scratch on CPU, reach 65.0% on FER2013 — roughly human-level on this benchmark.

The pipeline is the product

Detection, cropping, TTA and temporal smoothing matter as much as the architecture. Grad-CAM keeps it honest.

Evaluate per class, not just overall

61.5% macro F1 vs 65.0% accuracy shows the imbalance tax. Disgust and fear are where the work remains.

PyTorch · OpenCV · FER2013 · Grad-CAM · reproducible

Full code, weights, figures and this deck live in .

Questions?

Dataset: Goodfellow et al., FER2013 (2013) · Grad-CAM: Selvaraju et al. (2017)
Built with PyTorch, OpenCV and scikit-learn · 2026